

Assignment 2 - Statistics

Objective

To calculate the mean height of UK males from historical data and visualize the trend using Python.

Import Libraries

The required libraries are:

- Pandas for data handling
- Matplotlib for data visualization

```
import pandas as pd
import matplotlib.pyplot as plt
```

Creating the Dataset

The historical dataset contains the average height (in cm) of UK males from 1810 to 1980.

```
data = {
    "Year": [1810, 1820, 1830, 1840, 1850, 1860, 1870, 1880,
             1900, 1910, 1920, 1930, 1940, 1950, 1960, 1970, 1980],
    "Height": [169.7, 169.1, 166.7, 166.5, 165.6, 166.6, 167.2,
               168.0, 169.4, 170.9, 171.0, 173.9, 174.9,
               176.0, 176.9, 177.1, 176.8]
}

df = pd.DataFrame(data)

df
```

	Year	Height	
0	1810	169.7	
1	1820	169.1	
2	1830	166.7	
3	1840	166.5	
4	1850	165.6	
5	1860	166.6	
6	1870	167.2	
7	1880	168.0	
8	1900	169.4	
9	1910	170.9	
10	1920	171.0	
11	1930	173.9	
12	1940	174.9	
13	1950	176.0	
14	1960	176.9	
15	1970	177.1	
16	1980	176.8	

Next steps: [Generate code with df](#) [New interactive sheet](#)

Calculating the Mean Height

The average height is calculated using the `mean()` function.

```
mean_height = df["Height"].mean()

print(f"Mean Height: {mean_height:.2f} cm")
```

Mean Height: 170.96 cm

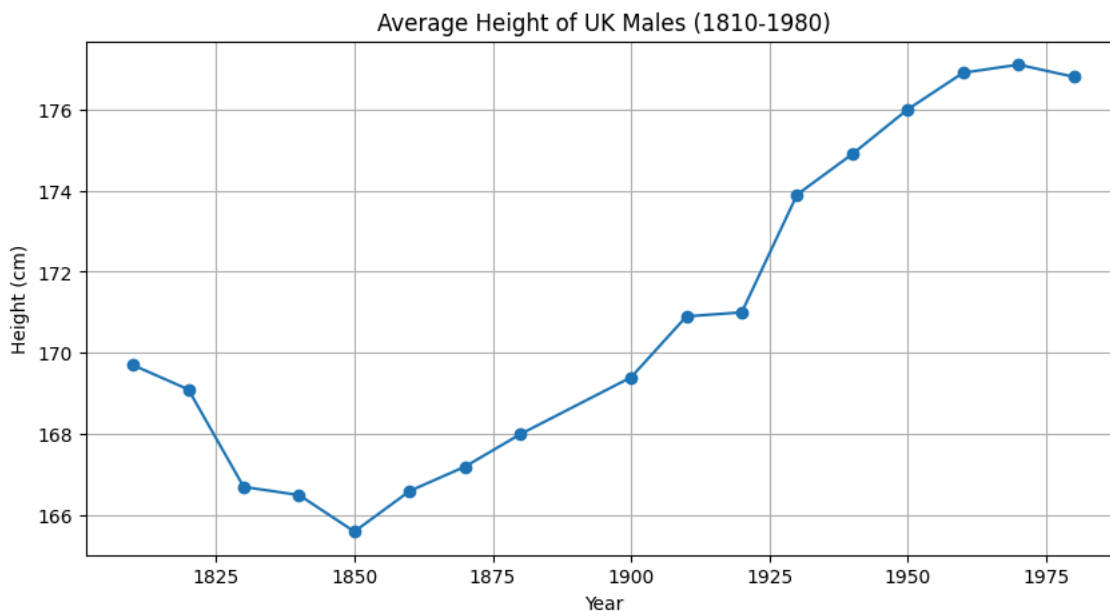
Data Visualization

The following line chart shows the trend in average height of UK males over the years.

```
plt.figure(figsize=(10,5))
plt.plot(df["Year"], df["Height"], marker='o')

plt.title("Average Height of UK Males (1810-1980)")
plt.xlabel("Year")
plt.ylabel("Height (cm)")
plt.grid(True)

plt.show()
```



Conclusion

The mean height of UK males in the given dataset is **170.96 cm**. The line graph shows that the average height gradually increased from 1810 to 1980. This trend may reflect improvements in nutrition, healthcare, and overall living conditions over time.

